(Please write your Exam Roll No.) Exam Roll No. .....

# END TERM EXAMINATION

**SECOND SEMESTER [B.TECH] JUNE 2024**

***Paper Code: BS-112*** ***Subject: Applied Mathematics-II***

***Time: 3 Hours*** ***Maximum Marks: 60***

***Note: Attempt any five questions including Q.No1 which is compulsory. Select one question from each unit.***

Q1 Attempt all of the following:- **(3x4=12)**

(a) Resolve $e^{\sin(x+iy)}$ into real and imaginary parts

(b) Find the image of infinite strip $\frac{1}{4} \le y \le \frac{1}{2}$. under the transformation $w = \frac{1}{z}$.

(c) Find the Laplace transform of $\frac{Cosat - Cosbt}{t}$.

(d) Using the method of Separation of variable , solve $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$

Where $(x, 0) = 6e^{-3x}$ .

**UNIT-I**

Q2 (a) Find the value of $C_1$ and $C_2$ such that the function **(6)**

$f(z) = x^2 + C_1y^2 - 2xy + i(C_2x^2 - y^2 + 2xy)$ is analytics. Also find $f'(z)$.

(b) Show that the function $u = \frac{1}{2}\log(x^2 + y^2)$ is harmonic . Find its harmonic Conjugate. **(6)**

Q3 (a) State Cauchy Integral formula and hence evaluate **(6)**

$\int_C \frac{3z^2 + z}{z^2 - 1} dz$ , where C is the circle $|z - 1| = 1$.

(b) Evaluate the Line Integral $\int_C z^2 dz$ , Where C is the boundary of a triangle with vertices $0, 0 + i, -1 + i$, Clockwise . **(6)**

**UNIT-II**

Q4 (a) Evaluate $\int_C \frac{z^2}{(z-1)^2(z+2)} dz$, Where C is $|z| = 3$ by using Cauchy residue theorem. **(6)**

(b) Evaluate Laurents series which represents the function **(6)**

$f(z) = \frac{z^2 - 1}{(z+2)(z+3)}$ When (i) $|z| < 2$, (ii) $2 < |z| < 3$ .

Q5 (a) Apply the calculus of residues to evaluate **(6)**

$\int_{-\infty}^{\infty} \frac{\cos x}{(x^2 + a^2)(x^2 + b^2)} dx$ , $a > b > 0$ .

(b) Let $f(z)$ be a bilinear transformation such that $f(\infty) = 1$, $f(i) = i$ , *and* $f(-i) = -i$

Find the image of unit dick (c) $\{z \epsilon c; |z| < 1\}$ under $(z)$ . **(6)**

**P.T.O.**

P-1/2
BS-112

(Please write your Enrolment No. immediately) Enrollment No .....

# Mid Term Examination

Second Semester, April ,2024

Paper Code: BS 112 Subject: Applied Mathematics -II

Time $1\frac{1}{2}$ Hrs. Max. Marks:30

**Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining .**

| Q | | Max. Marks | CO(s) |
|---|---|---|---|
| 1(a) | Convert the complex number $\frac{-16}{1+i\sqrt{3}}$ into Polar form . | 2.5 | 1 |
| (b) | Use Cauchy Integral Formula to evaluate $\int_c \frac{\text{Sin}\,\pi z^2 + \text{Cos}\,\pi z^2}{(z-1)(z-2)} dz$ ,where C is $\|z\|=3$ | 2.5 | 1 |
| (c) | Find Invariant Point in $w = \frac{3z-4}{z-1}$ | 2.5 | 2 |
| (d) | Determine the residue of function at each Pole. If, $f(z) = \frac{z^2}{(z-1)(z-2)^2}$ . | 2.5 | 2 |
| 2(a) | Find the Analytic function $f(z) = u + iv$ , $Given\ that\ v = e^x(\text{xsin}\,y + y\cos y)$ | 5 | 1 |
| (b) | Evaluate the Integral I = $\int_C \text{Re}\,(z^2)dz$ .From Z=0 to 2+4i along the Parabola $y = x^2$ | 5 | 1 |
| 3(a) | Show that $\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{(a^2-b^2)^{\frac{1}{2}}}$ , a> b>0 . Using Contour Integration. | 5 | 2 |
| (b) | Find the Laurent's expansion of $f(z) = \frac{1}{z(z^2-3z+2)}$ For the region $1 \leq \|z\| < 2$ . | 5 | 2 |
| 4(a) | Find the image of circle $x^2 + y^2 = 4y$ $Under\ the\ transformation\ w = \frac{1}{z}$ . | 5 | 2 |
| (b) | Discuss the analyticity of the function $f(z) = z\bar{z}$ at origin. | 5 | 1 |

*(Please write your Exam Roll No.)* *Exam Roll No. .....*

# END TERM EXAMINATION

## SECOND SEMESTER [B.TECH] JULY 2023

***Paper Code: BS-112*** ***Subject: Applied Mathematics-II***

***Time: 3 Hours*** ***Maximum Marks: 75***

***Note: Attempt five questions in all including Q.No. 1 which is compulsory. Select one question from each unit.***

Q1. (a) *Find all the point at which the following mapping is not Conformal* $w = \frac{z+\frac{1}{2}}{4z^2+2}$. **(2.5)**

(b) *Split the real and imaginary part of* $i^i$. **(2.5)**

(c). *Find the Laplace transform of* $t^2e^{-2t}$. **(2.5)**

(d) *Using half range sine series of function* $f(x) = 1$ *for* $0 < x < \pi$,

*prove that* $1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \ldots\ldots = \frac{\pi^2}{8}$ **(2.5)**

(e) *Taylor series expansion of* $\frac{1}{z-2}$ *in* $|z| < 1$ *is*..... **(2.5)**

(f) *Classify the type of PDE:* $\frac{\partial^2 u}{\partial x^2} + 2\frac{\partial^2 u}{\partial x \partial y} + 5\frac{\partial^2 u}{\partial y^2} = 0$, *whether it is parabolic, ellipticor hyperbolic?* **(2.5)**

**UNIT-I**

Q 2. (a) *Find all the values of* $(-1 + \sqrt{3}\, i)^{\frac{3}{2}}$. **(7)**

(b)*Verify that the function* $u(x, y) = x^3y - xy^3$ *is harmonic and find the harmonic conjugate of* $u(x, y)$ *to express the function* $f(z) = u + iv$ *as an analytic function.* **(8)**

Q 3. (a)*Evaluate the integral* $\oint_C \frac{e^{2z}}{(z-1)(z-2)} dz$ *along the curve* $C$, *where* $C$ *is a circle* $|z| = 3$. **(7)**

(b) *Integrate the function* $f(z) = \bar{z}$ *along the curve* $C$, *where* $C$ *is the square with vertices* $z = 0, 2, 2i, 2 + 2i$ **(8)**

**UNIT-II**

Q 4.(a) *Find the bilinear transformation or Mobious transformation which maps* $1, i, 1$ *of the* $z-$*plane onto* $1, i, -1$ *of the* $w-$*plane respectively. Also find the fixed points or invariant points.* **(8)**

**P.T.O.**

*(Please write your Exam Roll No.)* *Exam Roll No. .....*

# END TERM EXAMINATION

**SECOND SEMESTER [B.TECH] JULY 2023**

***Paper Code: ETMA-102*** ***Subject: Applied Mathematics-II***

***Time: 3 Hours*** ***Maximum Marks: 75***

***Note: Attempt five questions in all including Q.No1 which is compulsory. Select one question from each unit.***

**Q1).**

**a)** If $u = e^{xyz}$, show that $\frac{\partial^3 u}{\partial x\,\partial y\,\partial z} = (1 + 3xyz + x^2y^2z^2)e^{xyz}$. [4]

**b)** Form a partial differential equation of $z = (x^2 + a^2)(y^2 + b^2)$. [4]

**c)** Find Laplace transform of $f(t) = te^{-4t}\sin 3t$. [4]

**d)** Prove that the function sinhz is analytic and find its derivative. [4]

**e)** Check whether the vector field $F = 2xyz^3i + x^2z^3j + 3x^2yz^2k$ is irrotational. [4]

**f)** Evaluate $\int_0^1 \int_{e^x}^{e} \frac{1}{logy}\,dydx$, *by changing the order of integration.* [5]

**UNIT - I**

**Q2.**

**a)** If $x + y + z - u = 0,\ y + z - uv = 0,\ z - uvw = 0,$ show that $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2v$. [6]

**b)** Examine the function $f(x, y) = x^3 + y^3 - 12x - 3y + 20$ for extremum. [6.5]

**Q3.**

**a)** Find the maximum and minimum distances of the point $(1,2,-1)$ from the sphere $x^2 + y^2 + z^2 = 24$. [6.5]

**b)** If $V = r^m$, and $r^2 = x^2 + y^2 + z^2$, show that $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = m(m + 1)r^{m-2}$. [6]

**UNIT-II**

**Q4.**

**a)** Find Laplace transform of $\int_0^t \int_0^t \int_0^t (tsint)dtdtdt$. [6]

**b)** State convolution theorem and using it find inverse Laplace transform of $\frac{1}{s^2(s^2+a^2)}$ [6.5]

**Q5.**

**a)** Find Laplace transform of $f(t) = \frac{cos2t - cos3t}{t}$. [6]

**b)** Using Laplace transform solve: $y''' + 2y'' - y' - 2y = 0$, given $y(0) = 1,\ y'(0) = 2,$ $y''(0) = 2$. [6.5]

**UNIT - III**

**Q6.**

**a)** Find the bilinear transformation which maps the points $z = 1, -i, -1$ into the points $w = i, 0, -i$ [6]

**b)** Evaluate the integral $\int_0^{2\pi} \frac{d\theta}{1 - 2asin\theta + a^2}$, $0 < a < 1$. [6.5]

P.T.O.